Cairo University
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Von Neumann RISC Processor Design

Team 9

CMPS301

To: Eng. Sherif

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Instruction Format

Overview

- Fixed 32-bit instruction word
- Memory: 4*4KB (12-bit address space)
- Core fields:
    - opcode (5 bits)
    - rdst, rsrc1, rsrc2 (3 bits each)
    - imm/offset (up to 16 bits)
    - Jmpflags(2 bits)

Instruction Formats

F0 – No Operand

- Structure: opcode and all zeros
- Used by: NOP, HLT, SETC, RET, RTI

F1 – One Operand

- Register mirrored in rdst and rsrc1
- Used by: NOT, INC, IN, OUT, PUSH, POP

F2 – Register Operations

- 2-register (F2-2R): MOV, SWAP
- 3-register (F2-3R): ADD, SUB, AND

F3 – Immediate / Branch

- Branch (Addr12):
    - 12-bit target address
    - Early decode bits:
        - instr[16]: unconditional branch
        - instr[17]: conditional branch
- Immediate formats:
    - LDM, IADD, LDD, STD
- Interrupt format:
    - INT uses [1:0] index encoding

General Rules

- Immediate values stored in [15:0]
- Branch addresses stored in [11:0]
- Internal micro-ops (SWAP2, INT2, INT3) are not useable in assembler

Control Unit Design

Control Word

- **Execution & ALU**
  - ALU_OP
  - UPDATE_FLAGS
  - LOAD_FLAGS
- **Memory Control**
  - MEMR, MEMW
  - MEM_ADDRESS_SEL
  - MEM_WRITE_SEL
- **Register & I/O**
  - REG_WB_EN
  - OUTPUT_PORT_EN
- **Control Flow**
  - PC_WRITE_EN
  - COND_BRANCH
  - JMP_FLAG_SEL
- **Pipeline Control**
  - MULTICYCLE_STALL
  - MULTICYCLE_SEL
  - SWAP_2ND_CYCLE

Multicycle Sequencing

- Complex instructions broken into internal steps:
  - SWAP -> SWAP2
  - INT -> INT2 -> INT3
  - RTI -> RET
- Decode stage injects internal operations while stalling fetch

Priority Rules

- **PC-writing instructions have highest priority**
  - Stall other instructions
  - Flush pipeline on commit

Pipeline Design & Registers

Stage Responsibilities

Fetch

- Instruction fetch + next PC
- Early branch decoding and detection
- Branch prediction

Decode

- Instruction decode + register read
- Control signal generation
- Multicycle handling

Execute-1

- ALU operations
- Data forwarding applied

Execute-2

- Branch resolution, memory address calculation and branch misprediction detection

Memory

- Memory access
- Stack operations

Write-Back

- Register and flag updates

Pipeline Registers

- **IF_ID**
  - instr (32)
  - next_pc (32)
  - branch_prediction
- **ID_EX1**
  - Control word
  - Data:
    - register values (32-bit)
    - imm_offset (16)
    - register indices (3-bit)
    - next_pc (32)
- **EX1_EX2**
  - ALU outputs:
    - alu_result (32)
    - alu_flags (3)
  - Control signals
- **EX2_MEM**
  - Memory address/data (32-bit)
  - Flags and control signals
- **MEM_WB**
  - Write-back data (32-bit)
  - Flags (3-bit)
  - Register index

Pipeline Hazards

Data Hazards

- **Forwarding paths:**
    - EX1/EX2 → EX1
    - EX2/MEM → EX1
    - MEM/WB → EX1
- Forwarding to:
    - ALU inputs
    - Branch flags

Special Cases

- SWAP2 disables certain forwarding

Load-Use Hazard

- 1-cycle stall
- Up to 2 cycles if dependency persists

Structural Hazards

- Memory arbitration:
    - Memory stage gets priority
    - Fetch is stalled (FETCH_MEMORY_HAZARD)

Control Hazards

Misprediction Handling

- Flush:
    - IF/ID
    - ID/EX1
    - EX1/EX2
- Correct PC:
    - Target (if wrong-not-taken)
    - Next PC (if wrong-taken)

Special Control Cases

- PC-write instructions:
    - Stall pipeline
    - Flush on completion

5. Dynamic Branch Prediction

Predictor Type

- **2-bit counter**

Operation

In Fetch Stage

- Early detection using:
    - instr[16] -> unconditional branch
    - instr[17] -> conditional branch
- Predictor decides:
    - Taken / Not taken

In Execute-2 Stage

- Actual branch result computed
- Compared with prediction

Update Mechanism

- Counter updated based on correctness:
    - Strong/Weak Taken
    - Strong/Weak Not Taken

Benefits

- Reduces pipeline stalls
- Improves instruction throughput

Misprediction Cost

- Flush 3 pipeline stages
- Correct PC injected